

Cost-aware blocking: one-page strategy

Business entity matching | Source 1 versus Source 2 + Source 3

Decision: use rare, specific keys to avoid huge blocks; retrieve a short ranked list with the existing TF-IDF + BM25 channels; add a small rescue route for difficult names. Cap the final, deduplicated list at 50 candidates per Source-1 row for the first pilot, then adjust from measured recall and runtime.

Why the cost problem is real

Source 1 has 2,206,821 rows and the two target sources each have about 5.3 million. Comparing every Source-1 row with every target would mean about 23 trillion pairs. Even 50 candidates each still means up to 110.3 million ML-scored pairs; process them in batches. A broad key such as “road” can also make retrieval itself expensive before the top 50 are selected.

Recommended five-stage pipeline

1. Clean, keep Unicode. Use normalized names and addresses, retain original text, and add transliteration as a second search form. Do not make it the only representation.
2. Cheap, rare keys. Within a valid country, use exact/rare name tokens combined with locality, house number, or rare address tokens. Skip or cap very common postings. For missing country, allow a bounded fallback.
3. Ranked search. Run name and address character TF-IDF plus BM25; deduplicate results. Add transliteration search for applicable scripts. Limit expensive posting scans as well as output candidates.
4. Targeted rescue. For short/initial names or weak name retrieval, search rare address + number/locality or transliteration paths; allocate a small extra budget. Do not stop after a single apparent match.
5. Cap and score. Keep the best 50 unique target IDs per Source-1 row in the pilot. Generate pair features and run the saved ML model only on this bounded list.

How to choose the final budget (not a guessed winner)

On a 20,000-query pilot, compare the current retriever at K=20/50/100 with the proposed keys and rescue route. Record true-link recall@K, per-query coverage, p95/p99 candidates, posting entries scanned, elapsed time and peak RAM. Then choose the cheapest setting that meets the recall target; test unchanged on a grouped holdout. Do not insert true matches into candidate lists or tune the classifier threshold on holdout.

Important: the supplied EDA key-sharing rates (e.g., 98.54% for name-trigram OR address-number) are uncapped possibilities on sampled true pairs, not recall@50, model F0.5 or proven runtime. The 50 cap and posting ceiling are starting settings, not measured optima.

Data check before final claims: current Source-3 file counts 5,274,632 rows, while an earlier EDA said 5,285,603. A partial index also reported 8,193 unknown target IDs. Reconcile complete file counts and ground-truth IDs first; do not treat a partial-index count as a confirmed dataset defect.